

a**Form the target**

Soft-SFA supplies three local forces.

Slowness

$$r_t^{\text{slow}} = I_t(z_t - p_t)$$

Variance preservation

$$r_t^{\text{var}} = -\frac{\lambda_V}{\lambda_S} g_t \odot (z_t - \mu)$$

Decorrelation

$$r_t^{\text{cov}} = \frac{\lambda_C}{2\lambda_S} A(z_t - \mu)$$

$$\hat{z}_t = \text{sg}\left(z_t - r_t^{\text{slow}} - r_t^{\text{var}} - r_t^{\text{cov}}\right)$$

b**Define the neuron state**

Map the target residual to preactivation.

Base state

$$b_t = J_{\phi,t}^T (z_t - \hat{z}_t)$$

one inhibitory pass

Neuron state

$$s_t = b_t - \eta \kappa M b_t$$

A postsynaptic state retained only for the current local update.

c**Update local synapses**

The same state supplies both updates.

Feedforward synapse

$$\nabla_{W_{ji}} \tilde{L}_a \propto s_{t,j} x_{t,i}$$

Lateral synapse

$$\Delta M_{jk} \propto -s_{t,j} s_{t,k}$$

Endpoint activities determine each change; no error arrives from a later layer.

Local in time: fixed detached state

Local in space: endpoint signals and same-layer inhibition